

Supplementary material

Dynamics of temperature and pressure in air intake environment simulation system: The core dynamics of air intake system involve mixing high- and low-temperature airflows to reproduce desired flight conditions. By regulating two control valves, the combined airflow achieves the target temperature and pressure. For modeling purposes, the intake system is simplified into an equivalent dual-intake, single-exhaust cavity, as illustrated in Fig. 1.

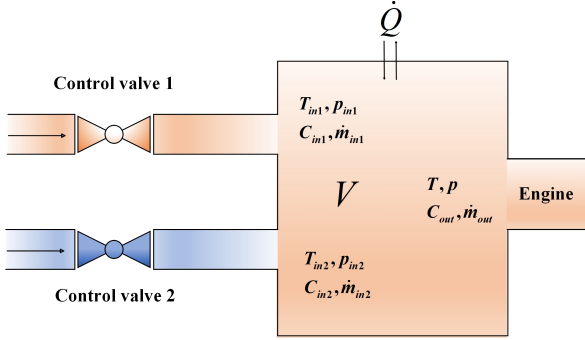


Fig 1 Schematic diagram of cavity structure for the intake air environment simulation system.

In Fig. 1, the basic physical variables T , C , p , and $\dot{m}$ denote the temperature, average flow velocity, pressure, and mass flow rate, respectively. Regarding the subscripts, $in1$ and $in2$ indicate the high-temperature and low-temperature intake pipelines, respectively, while the absence of a subscript (or the subscript out) refers to the exhaust of the mixer cavity. Additionally, $\dot{Q}$ represents the heat transfer through the cavity wall, and V is the volume of the mixer cavity. This letter employs a cavity model to describe the gas dynamics within the flight environment simulation system. It is assumed that the internal airflow velocity changes slowly, and that the exhaust gas states are identical to the internal states of the cavity. The detailed derivation of the temperature and pressure differential equations is provided in [1]. Applying these results yields

$$\begin{aligned} \frac{dT}{dt} = & \frac{T}{p} \frac{R}{V(c_p - R)} [-(\dot{m}_{in1} + \dot{m}_{in2} - \dot{m}_{out})(h_{out} - RT)] \\ & + \frac{T}{p} \frac{R}{V(c_p - R)} \left[\dot{Q} + \left(h_{in1} + \frac{C_{in1}^2}{2} \right) \dot{m}_{in1} \right. \\ & \left. + \left(h_{in2} + \frac{C_{in2}^2}{2} \right) \dot{m}_{in2} - \left(h_{out} + \frac{C_{out}^2}{2} \right) \dot{m}_{out} \right], \end{aligned} \quad (1)$$

$$\begin{aligned} \frac{dp}{dt} = & \left[\left(\frac{RT}{V} - \frac{R}{V(c_p - R)} (h_{out} - RT) \right) (\dot{m}_{in1} + \dot{m}_{in2} - \dot{m}_{out}) \right] \\ & + \frac{R}{V(c_p - R)} \left[\dot{Q} + \left(h_{in1} + \frac{C_{in1}^2}{2} \right) \dot{m}_{in1} \right. \\ & \left. + \left(h_{in2} + \frac{C_{in2}^2}{2} \right) \dot{m}_{in2} - \left(h_{out} + \frac{C_{out}^2}{2} \right) \dot{m}_{out} \right]. \end{aligned} \quad (2)$$

In Eqs. (1) and (2), h_{in1} , h_{in2} , and h_{out} denote the enthalpies of the high-temperature intake, low-temperature intake, and exhaust of air intake system cavity, respectively. Additionally, c_p is the specific heat at constant pressure, and R is the gas constant of the air within the cavity.

Model of control valve: As the actuator of air intake environment simulation system, the control valve's dynamic characteristics directly dictate the response speed and precision of airflow regulation. In practical engineering, the control valve and its associated electro-hydraulic servo drive system involve complex mechanical, electrical, and hydraulic multi-domain coupling. Consequently, their exact mechanistic models often present as complex high-order nonlinear systems. By combining theoretical analysis with experimental data identification, the dynamic response of the valve actuator can be approximated as a first-order inertial element

[2].

$$V_\theta = \frac{K_\theta}{T_\theta s + 1} u, \quad (3)$$

where K_θ is the proportional gain, and T_θ is the time constant reflecting the physical delay of the valve actuation. Governed by these parameters, the control signal u translates into the valve opening V_θ . Furthermore, by incorporating the fluid dynamic throttling effect [3, 4], the mapping relationship between the mass flow rates of dual intake control valves and their physical openings is established as

$$\dot{m}_{in1} = \varphi_0 A_0 \sqrt{2\rho_1 p_1} V_{\theta,1}, \quad \dot{m}_{in2} = \varphi_0 A_0 \sqrt{2\rho_2 p_2} V_{\theta,2}, \quad (4)$$

where $\dot{m}_{in1}$ and $\dot{m}_{in2}$ denote the mass flow rates through the hot and cold flow valves, respectively. $V_{\theta,1}$ and $V_{\theta,2}$ represent their real-time openings. A_0 is the flow area of the plug valve. ρ_1 and ρ_2 are the densities of the two gas streams, while p_1 and p_2 are their respective upstream pressures. Lastly, φ_0 is the dimensionless flow coefficient characterizing the impact of the flow channel design on flow efficiency.

Mathematical model of air intake environment simulation system: Substituting the flow-valve relationships into the thermodynamic dynamics (1) (2) yields the following affine nonlinear model

$$\dot{p} = f_p + b_{11} V_{\theta,1} + b_{12} V_{\theta,2}, \quad (5)$$

$$\dot{T} = f_T + b_{21} V_{\theta,1} + b_{22} V_{\theta,2}, \quad (6)$$

where f_p , f_T , b_{11} , b_{12} , b_{21} , and b_{22} denote the input coefficients.

$$f_p = - \left[\frac{RT}{V} - \Phi_R \Phi_B + \Phi_R \left(h_{out} + \frac{C_{out}^2}{2} \right) \right] \dot{m}_{out}, \quad (7)$$

$$b_{11} = \left[\frac{RT}{V} - \Phi_R \Phi_B + \Phi_R \left(h_{in1} + \frac{C_{in1}^2}{2} \right) \right] \varphi_0 A_0 \sqrt{2\rho_1 p_1}, \quad (8)$$

$$b_{12} = \left[\frac{RT}{V} - \Phi_R \Phi_B + \Phi_R \left(h_{in2} + \frac{C_{in2}^2}{2} \right) \right] \varphi_0 A_0 \sqrt{2\rho_2 p_2}, \quad (9)$$

$$f_T = \frac{T}{p} \Phi_R \left[\Phi_B - \left(h_{out} + \frac{C_{out}^2}{2} \right) \right] \dot{m}_{out}, \quad (10)$$

$$b_{21} = -\frac{T}{p} \Phi_R \left[\Phi_B - \left(h_{in1} + \frac{C_{in1}^2}{2} \right) \right] \varphi_0 A_0 \sqrt{2\rho_1 p_1}, \quad (11)$$

$$b_{22} = -\frac{T}{p} \Phi_R \left[\Phi_B - \left(h_{in2} + \frac{C_{in2}^2}{2} \right) \right] \varphi_0 A_0 \sqrt{2\rho_2 p_2}, \quad (12)$$

where the expressions for Φ_R and Φ_B are given as

$$\Phi_R = \frac{R}{V(c_p - R)}, \quad \Phi_B = h_{out} - RT. \quad (13)$$

Extended State Observer: The temperature and pressure systems share an identical second-order structure. Therefore, the following ESO is designed as

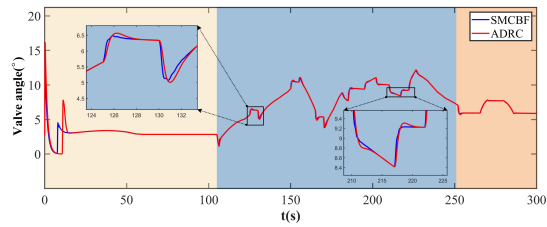
$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 - \beta_1 (\hat{x}_1 - x_1) \\ \dot{\hat{x}}_2 = \hat{x}_3 + U - \beta_2 (\hat{x}_1 - x_1) \\ \dot{\hat{x}}_3 = -\beta_3 (\hat{x}_1 - x_1) \end{cases}, \quad (14)$$

where $\hat{x}_1$, $\hat{x}_2$, and $\hat{x}_3$ represent the estimated values of the controlled variable, its derivative, and the system disturbance, respectively. The parameters β_1 , β_2 , and β_3 are determined by the ESO bandwidth ω , satisfying $\beta_1 = 3\omega$, $\beta_2 = 3\omega^2$, and $\beta_3 = \omega^3$.

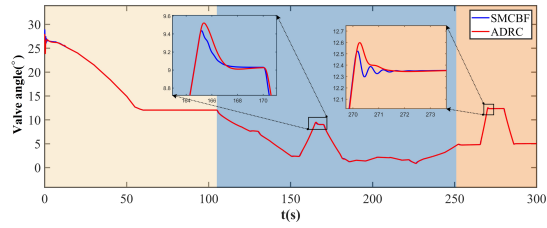
Valve response: As seen in Figs. 2, valves $V_{\theta,1}$ and $V_{\theta,2}$ exhibit smaller oscillations under SMCBF control, which implies a reduction in mechanical wear.

References

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(a) Valve opening $V_{\theta,1}$



(b) Valve opening $V_{\theta,2}$

Fig 2 Valve response

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